Learning with Less: Training ML Models with Incomplete Human Feature Sets

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Introduction

Machine Learning (ML) models typically get trained, validated and tested on a fixed set of input variables for all predictions. This can overlook valuable and individualized knowledge that human users know about themselves and therefore limit accuracy. For example, an ML model predicting the likelihood of college success might miss important personal factors, such as their studying habits, sleeping schedule and diet that only the individual can provide.

While studies have shown that incorporating user-provided features (or human features) may improve prediction accuracy. An approach has been developed for selecting which features to query from users and integrate into predictions, but this approach assumes all features are available at train time. Real-world scenarios often involve missing or selectively available features, particularly when human features are expensive to collect or when feature availability varies across predictions.

We study how to optimally train machine learning models when only a subset of features will be available at train time. We have different strategies for data collection including various strategies for random selection, uncertainty-based approaches, and heuristic methods. We run our experiments across three semi-realistic datasets to examine two key questions: (1) what training strategy works best for the existing approach to integrating user features at test time, and (2) whether the training strategy needs to be matched to the testing strategy.

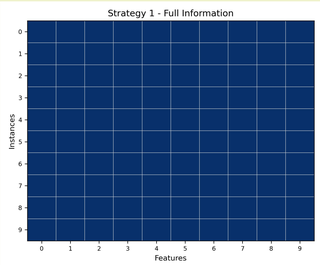
Methods

We formalize the problem as feature masking: each time we train the model, we will pass a binary mask of shape *(number of instances) x (number of human features)* into the model where 1 means the data is available and will be included in training/testing and 0 means the data is unavailable and can’t be used.

We evaluate our approaches using three datasets: a toy dataset which is generated to validate our methods work in principle, and two datasets (covertype and recipe) to test if the heuristic approach works when we have access to full training data.

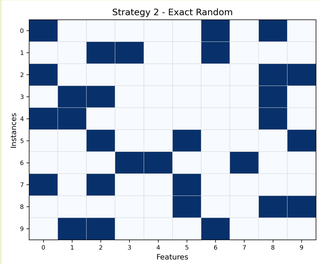
Main idea: Thought experiment: if we know exactly how we’ll use this model, does training it that way make a big difference?

Masking strategy



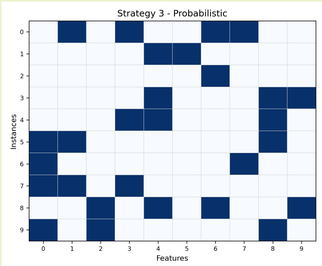
Strategy 1: Full information

- All features are available for all instances
- Mask consists entirely of 1’s
- The upper-bound baseline



Strategy 2: Exact random selection

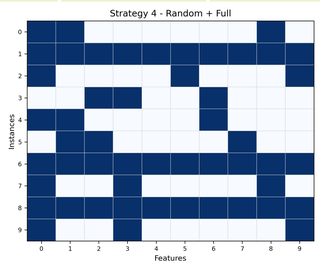
- Each instance provides exactly n features, where n is predetermined
- Each row contains exactly n ones, with positions randomly distributed



Strategy 3: Probabilistic selection

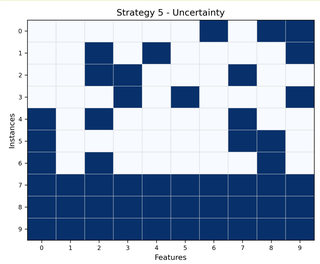
- Each feature is available with probability p

On average, each instance contains approximately *p × (total number of features)* available features



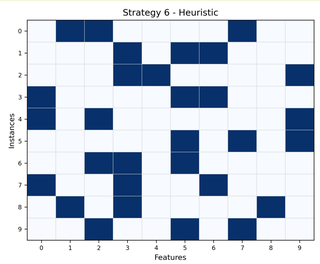
Strategy 4: Combination of random-full selection

- A subset of instances provides complete information (all features available)
- Remaining instances follow Strategy 2 or Strategy 3
- Complete information instances are selected randomly
- Simulates scenarios where some users provide comprehensive data and others don’t



Strategy 5: Uncertainty selection

- Similar to Strategy 4, but complete information instances are selected strategically: iteratively select rows with highest uncertainty and include them in the next iteration.



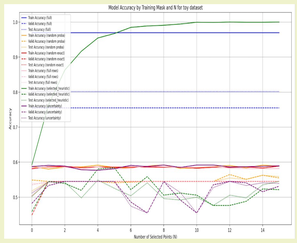
Strategy 6: Heuristic selection

- K features are selected intelligently (using an algorithm) for each row. This is how the model will be used in practice. Doing this selection requires training data (it gets circular–we need the data before we can figure out how to get this data) so this is another upper bound.

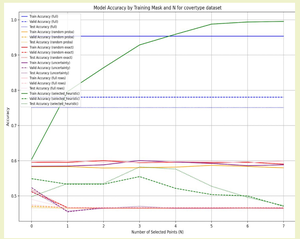
Experimental design

- **Training masks:** Vary the feature selection strategy during training:
  - **Full mask:** all features included
  - **Random exact:** exactly n features selected per instance
  - **Random probability p:** each feature included with independent probability p
  - **Combination of random– full selection:** exactly n features selected per instances or each feature included with independent probability p
  - **Heuristic selected:** features chosen based on heuristic criteria
  - **Uncertainty-based:** features selected based on prediction uncertainty
- **Testing masks:** full data mask
- **Parameters:** Number of features n, selection probability p, and mask types varied to simulate selective feature availability
- **Comparison:** All strategies are compared under controlled conditions with identical numbers of available features (same number of 1’s in masks) at each experimental stage to ensure fairness and consistency.

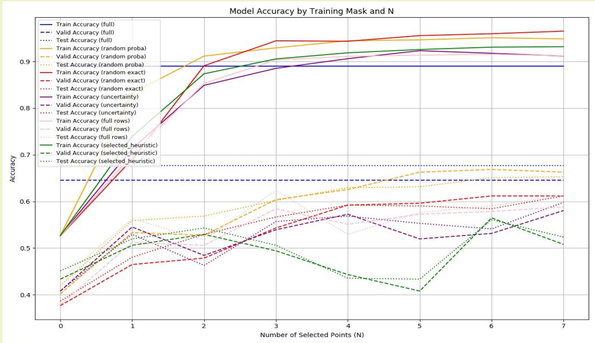
Results



Graph 1. Toy data set



Graph 2. Covertype data set



Graph 3. Recipe data set

Key findings

- Random feature selection often outperforms complex heuristic methods when full test data is available.
- The result is consistent (1) datasets; (2) how much training data you get (how many 1’s in the mask)

Future work

- Continue investigating research questions:
  - If a selected strategy is used at test time, what is the optimal training strategy?
  - Is there a best training strategy independent of the test strategy, or is it most important to match them?
- Test a wider range of uncertainty-based strategy for feature selection.
- Apply the framework to **real-world datasets** where selective feature availability reflects user input or system constraints.

Acknowledgements

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References

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